

# Understanding Analysis, 2nd Edition

## Chapter 4 - Section 4.5 - The Intermediate Value Theorem

1. Show how the IVT follows as a corollary to Theorem 4.5.2.

**Solution:** Theorem 4.5.2: continuous function preserves connected sets.

Definition of connected sets: A set  $E \subseteq \mathbf{R}$  is connected iff whenever  $a < c < b$  with  $a, b \in E$ , then  $c \in E$  as well.

In  $\mathbf{R}$ , connected sets coincide precisely with the collection of intervals (page 104).

Then IVT follows pretty much by definition. A closed interval  $[a, b]$  is connected. Let  $f$  be continuous, then  $f([a, b])$  is connected. By definition of connectedness,  $f$  contains everything in  $[f(a), f(b)]$ .

2. Example or prove impossible.
  - (a) A continuous function defined on an open interval with range equal to a closed interval.
  - (b) A continuous function defined on a closed interval with range equal to an open interval.
  - (c) A continuous function defined on an open interval with range equal to an unbounded closed set different from  $\mathbf{R}$ .
  - (d) A continuous function defined on all of  $\mathbf{R}$  with range equal to  $\mathbf{Q}$ .

**Solution:**

(a)  $\sin(1/x)$  is continuous on  $(0, 1)$  with range  $[-1, 1]$ .

(b) For unbounded interval, yes.  $f(x) = x$  is continuous on  $\mathbf{R}$  and its range is also  $\mathbf{R}$  (recall that  $\mathbf{R}$  is both open and closed).

For unbounded interval, impossible. A closed bounded interval is compact and continuous functions preserve compactness (Theorem 4.4.1).

(c) With unbounded interval:  $x^2$  is continuous on  $\mathbf{R}$  with range  $[0, \infty)$ .

With bounded interval:  $\tan^2 x$  is continuous on  $(\frac{\pi}{2}, \frac{\pi}{2})$  with range  $[0, \infty)$ .

(d) Impossible. We'll prove it by contradiction. Suppose there exists such a function. Since the range is  $\mathbf{Q}$ ,  $f$  is not constant. Choose  $x \neq y$  such that  $f(x) \neq f(y)$  and wlog  $f(x) < f(y)$ . By IVT, the range of  $f$  must contain everything in  $[f(x), f(y)]$ . But this implies  $f$  contains an irrational, since between any two rationals there is an irrational, contradicting the assumption that the range is  $\mathbf{Q}$ .

3. A function  $f$  is *increasing* on  $A$  if  $f(x) \leq f(y)$  for all  $x < y$  in  $A$ . Show that if  $f$  is increasing on  $[a, b]$  and satisfies the intermediate value property (Definition 4.5.3), then  $f$  is continuous on  $[a, b]$ .

**Solution:** To prove that  $f$  is continuous at  $a \leq c \leq b$ , we need to show that given  $\epsilon > 0$ , there exists  $\delta > 0$  such that  $|x - c| < \delta \implies |f(x) - f(c)| < \epsilon$ .

We'll prove left continuity: there exists  $\delta_a$  such that for  $x < c$  and  $c - x < \delta_a$ ,  $f(c) - f(x) < \epsilon$ . The proof for  $x > c$  is similar and will result in a  $\delta_b$ . Then choose  $\delta = \min(\delta_a, \delta_b)$ .

By monotonicity,  $f(a) \leq f(c)$  and if  $a < x < c$  then  $f(a) \leq f(x) \leq f(c)$ . By intermediate value property,  $f$  takes on all values between  $f(a)$  and  $f(c)$  so there is an  $a < x < c$  such that  $f(c) - f(x) < \epsilon$ . Choose  $\delta_a = c - x$ . This works for all  $x < t < c$  as well because monotonicity ensures that  $f(c) - f(t) \leq f(c) - f(x) < \epsilon$ .

4. Let  $g$  be continuous on an interval  $A$  and let  $F$  be the set of points where  $g$  fails to be one-to-one; that is,

$$F = \{x \in A : f(x) = f(y) \text{ for some } y \neq x \text{ and } y \in A\}.$$

Show  $F$  is either empty or uncountable.

**Solution:** At first I thought the problem statement must be wrong. Take something like  $\sin(x)$ . Within an interval there's only a finite number of points where  $\sin(x) = 1$ , so it's neither empty nor uncountable. But then I realized it's not only 1 that's repeating, the intervals around  $\sin(x) = 1$  must be repeating as well and an interval is uncountable. That's what we'll prove;  $F$  is either empty or contains repeating intervals of output values.

If  $g$  never fails to be one-to-one then  $F$  is trivially empty. Otherwise there exists  $x \neq y$  such that  $g(x) = g(y)$ .

If for all  $x < c < y$ ,  $g(c) = g(x) = g(y)$ , then  $[x, y] \subseteq F$  so  $F$  is uncountable.

Otherwise, let  $g(c) \neq g(x)$  and wlog  $g(c) > g(x)$ . By IVT there exists a subset  $S_1 \subseteq (x, c)$  such that  $g(S_1) = (g(x), g(c))$ . Since  $g(S_1)$  is an interval,  $S_1$  must be uncountable.

Similarly, there exists a subset  $S_2 \subseteq (c, y)$  such that  $g(S_2) = (g(y), g(c))$ .

Since  $g(x) = g(y)$  then  $g(S_1) = g(S_2)$ , hence  $S_1, S_2 \subseteq F$  and  $F$  is uncountable. Note that  $S_1 \cap S_2 = \emptyset$  so they're different set of points and can be included in  $F$ .

5. (a) Finish the proof of the Intermediate Value Theorem using the Axiom of Completeness started previously.
- (b) Finish the proof of the Intermediate Value Theorem using the Nested Interval Property started previously.

**Solution:**

- (a) The partial proof from the book is as follow. Let  $f$  be continuous and consider the special case  $f(a) < 0 < f(b)$ . We'll show that  $f(c) = 0$  for some  $c \in (a, b)$ .  
Let

$$K = \{x \in [a, b] : f(x) \leq 0\}.$$

Notice that  $K$  is bounded above by  $b$  and  $a \in K$  so  $K$  is not empty. Thus we may appeal to AoC to assert that  $s = \sup K$  exists.

We'll first prove that  $f(s) \leq 0$ . Since  $s$  is a supremum,  $s$  is a limit point of some sequence  $(x_n) \in K$ . By continuity,  $f(x_n) \rightarrow f(s)$ . Since  $f(x_n) \leq 0$ , then  $f(s) \leq 0$ .

Next we need to prove that  $f(s) \geq 0$ . The idea is that since  $s$  is a supremum, it's sort of the "rightmost" point where  $f$  is  $\leq 0$ . But if  $f(s) < 0$ , since everything to the right of  $s$  is  $> 0$ , there'd be a discontinuity; a jump that skips 0. Or in other words,  $f(s)$  is not continuous from the right side.

Formally, we prove by contradiction. First we note that since  $s = \sup K$ , if  $x > s$  then  $x \notin K$  hence  $f(x) > 0$ .

Now let  $\epsilon = |f(s)|$ . By continuity, there exists a  $\delta < (b - s)$  that works (the bound  $(b - s)$  is just a trivial detail to ensure we stay within  $[a, b]$ ). Choose  $x$  such that  $x - s < \delta$ . Since  $f$  is defined on  $[a, b]$ ,  $f(x)$  is defined.

We've asserted that  $f(x) > 0$ . But since  $f(s) < 0$ , we have  $f(x) - f(s) \geq |0 - f(s)| = \epsilon$ , contradicting continuity of  $f$ .